

**Hugging Face is an AI company and open-source platform that provides tools and pre-trained models for Natural Language Processing (NLP), computer vision, and machine learning.**

**It allows you to use powerful AI models (like BERT, GPT) with just a few lines of code.**

### 1. Transformers Library

Main library: Transformers (Hugging Face library)

Provides pre-trained models like: BERT GPT

Tasks supported: Text classification Translation Question answering Summarization

### 2. Model Hub A huge collection of pre-trained models

Thousands of models shared by researchers & developers You can:

Download models

Upload your own models

### 3. Datasets Library

Provides ready-to-use datasets

### This installs:

Transformers (Hugging Face library) Hugging Face datasets library

```
!pip install transformers datasets
```

```
Requirement already satisfied: transformers in /usr/local/lib/python3.12/dist-packages (5.0.0)
Requirement already satisfied: datasets in /usr/local/lib/python3.12/dist-packages (4.0.0)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from transformers) (3.25.2)
Requirement already satisfied: huggingface-hub<2.0,>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (1.8.0)
Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.12/dist-packages (from transformers) (2.0.2)
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Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.12/dist-packages (from transformers) (2025.11.3)
Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (0.22.2)
Requirement already satisfied: typer-slim in /usr/local/lib/python3.12/dist-packages (from transformers) (0.24.0)
Requirement already satisfied: safetensors>=0.4.3 in /usr/local/lib/python3.12/dist-packages (from transformers) (0.7.0)
Requirement already satisfied: tqdm>=4.27 in /usr/local/lib/python3.12/dist-packages (from transformers) (4.67.3)
Requirement already satisfied: pyarrow>=15.0.0 in /usr/local/lib/python3.12/dist-packages (from datasets) (18.1.0)
Requirement already satisfied: dill<0.3.9,>=0.3.0 in /usr/local/lib/python3.12/dist-packages (from datasets) (0.3.8)
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (from datasets) (2.2.2)
Requirement already satisfied: requests>=2.32.2 in /usr/local/lib/python3.12/dist-packages (from datasets) (2.32.4)
```

```

Requirement already satisfied: xxhash in /usr/local/lib/python3.12/dist-packages (from datasets) (3.6.0)
Requirement already satisfied: multiprocess<0.70.17 in /usr/local/lib/python3.12/dist-packages (from datasets) (0.70.16)
Requirement already satisfied: fsspec<=2025.3.0,>=2023.1.0 in /usr/local/lib/python3.12/dist-packages (from fsspec[http]<=2025.3.0,>=2023.1.0->datasets) (2025.3.0)
Requirement already satisfied: aiohttp!=4.0.0a0,!4.0.0a1 in /usr/local/lib/python3.12/dist-packages (from fsspec[http]<=2025.3.0,>=2023.1.0->datasets) (3.13.4)
Requirement already satisfied: hf-xet<2.0.0,>=1.4.2 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0->transformers) (1.4.2)
Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0->transformers) (0.28.1)
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Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.2->datasets) (3.11)
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Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas->datasets) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas->datasets) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas->datasets) (2025.3)
Requirement already satisfied: aiohappyeyeballs>=2.5.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]<=2025.3.0,>=2023.1.0->d
Requirement already satisfied: aiosignal>=1.4.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]<=2025.3.0,>=2023.1.0->datasets)
Requirement already satisfied: attrs>=17.3.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]<=2025.3.0,>=2023.1.0->datasets) (
Requirement already satisfied: frozenlist>=1.1.1 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]<=2025.3.0,>=2023.1.0->dataset
Requirement already satisfied: multidict<7.0,>=4.5 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]<=2025.3.0,>=2023.1.0->datas
Requirement already satisfied: propcache>=0.2.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]<=2025.3.0,>=2023.1.0->datasets)
Requirement already satisfied: yarl<2.0,>=1.17.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]<=2025.3.0,>=2023.1.0->dataset
Requirement already satisfied: anyio in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.3.0->transformers) (4.13.0)
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Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.3.0->transformers) (0
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas->datasets) (1.17.0)
Requirement already satisfied: click>=8.2.1 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,>=1.3.0->transformers) (8.3.1)
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Requirement already satisfied: rich>=12.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,>=1.3.0->transformers) (13.9.4)
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,>=1.3.0->transformers) (0.0.4)
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer->huggingface-hub<2.0,>=1.3.0->transformers) (4
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer->huggingface-hub<2.0,>=1.3.0->transformers) (
Requirement already satisfied: mdurl~0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0->rich>=12.3.0->typer->huggingface-hub<2.0,>=1.3.0->tran

```

## ✦ Import Libraries

```
from transformers import pipeline
```

## ✦ Sentiment Analysis

This line automatically:

Downloads model Loads tokenizer Initializes pipeline

```
classifier = pipeline("sentiment-analysis")
```

```
result = classifier("Hugging Face is very easy to use!")
print(result)
```

No model was supplied, defaulted to distilbert/distilbert-base-uncased-finetuned-sst-2-english and revision 714eb0f.  
Using a pipeline without specifying a model name and revision in production is not recommended.  
/usr/local/lib/python3.12/dist-packages/huggingface\_hub/utils/\_auth.py:94: UserWarning:  
The secret `HF\_TOKEN` does not exist in your Colab secrets.  
To authenticate with the Hugging Face Hub, create a token in your settings tab (<https://huggingface.co/settings/tokens>), set it as secret in your Google Colab and restart the notebook.  
You will be able to reuse this secret in all of your notebooks.  
Please note that authentication is recommended but still optional to access public models or datasets.

```
warnings.warn(
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.

config.json: 100%                               629/629 [00:00<00:00, 6.97kB/s]

model.safetensors: 100%                         268M/268M [00:05<00:00, 418MB/s]

Loading weights: 100%                          104/104 [00:00<00:00, 244.45it/s, Materializing param=pre_classifier.weight]

tokenizer_config.json: 100%                     48.0/48.0 [00:00<00:00, 818B/s]

vocab.txt:      232k/? [00:00<00:00, 1.99MB/s]
[{'label': 'POSITIVE', 'score': 0.9988467693328857}]
```

## Dataset Importing

```
from datasets import load_dataset
dataset = load_dataset("imdb")
print(dataset)
```

```

README.md:      7.81k/? [00:00<00:00, 149kB/s]

plain_text/train-00000-of-00001.parquet: 100%      21.0M/21.0M [00:00<00:00, 104MB/s]

plain_text/test-00000-of-00001.parquet: 100%      20.5M/20.5M [00:01<00:00, 102MB/s]

plain_text/unsupervised-00000-of-00001.p(...): 100%      42.0M/42.0M [00:02<00:00, 39.5MB/s]

Generating train split: 100%      25000/25000 [00:00<00:00, 35977.71 examples/s]

Generating test split: 100%      25000/25000 [00:00<00:00, 62874.05 examples/s]

Generating unsupervised split: 100%      50000/50000 [00:01<00:00, 54816.10 examples/s]

DatasetDict({
  train: Dataset({
    features: ['text', 'label'],
    num_rows: 25000
  })
  test: Dataset({
    features: ['text', 'label'],
    num_rows: 25000
  })
  unsupervised: Dataset({
    features: ['text', 'label'],
    num_rows: 50000
  })
})

```

## ✓ Part 2: Using BERT Model

### Load Tokenizer & Model

```

from transformers import AutoTokenizer, AutoModel

tokenizer = AutoTokenizer.from_pretrained("bert-base-uncased")
model = AutoModel.from_pretrained("bert-base-uncased")

```

### Tokenization

```

text = "Transformers are powerful models"

inputs = tokenizer(text, return_tensors="pt")
print(inputs)

```

```

outputs = model(**inputs)

# CLS token embedding

```

```
cls_embedding = outputs.last_hidden_state[:, 0, :]  
print(cls_embedding.shape)
```

## ✓ Text Classification Using Hugging Face With Sample Data

Text → Tokenizer → BERT → CLS Embedding → Naive Bayes → Prediction

```
from transformers import AutoTokenizer, AutoModel  
import torch  
import numpy as np  
from sklearn.naive_bayes import GaussianNB  
from sklearn.model_selection import train_test_split
```

```
tokenizer = AutoTokenizer.from_pretrained("bert-base-uncased")  
model = AutoModel.from_pretrained("bert-base-uncased")
```

```
texts = [  
    "I love this product",  
    "This is amazing",  
    "I am very happy",  
    "I hate this",  
    "This is bad",  
    "Very disappointing"  
]  
  
labels = [1, 1, 1, 0, 0, 0]
```

```
def get_embedding(text):  
    inputs = tokenizer(text, return_tensors="pt", truncation=True, padding=True)  
  
    outputs = model(**inputs)  
  
    # CLS embedding  
    cls_embedding = outputs.last_hidden_state[:, 0, :]  
  
    return cls_embedding.detach().numpy()[0]
```

```
X = np.array([get_embedding(t) for t in texts])  
y = np.array(labels)  
  
print("Feature shape:", X.shape)
```

```
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)
```

```
clf = GaussianNB()  
clf.fit(X_train, y_train)
```

```
y_pred = clf.predict(X_test)  
  
print("Predictions:", y_pred)  
print("Actual:", y_test)
```

```
test_text = "This product is really good"  
  
test_emb = get_embedding(test_text).reshape(1, -1)  
  
prediction = clf.predict(test_emb)  
  
print("Prediction:", "Positive" if prediction[0] == 1 else "Negative")
```

Start coding or [generate](#) with AI.